

Cloud Storage Security

1. Create a new storage bucket in your chosen cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) named 'my-music-uploads' and upload any 2 MP3 files to it.

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1. Log in to the AWS Management Console.
 2. Open Amazon S3.
 3. Click Create Bucket.
 4. Enter the bucket name:
my-music-uploads
Note: S3 bucket names must be globally unique. If my-music-uploads is already taken, use a unique name such as:
my-music-uploads-deep-20260718
 5. Select your preferred AWS Region (for example, Asia Pacific (Mumbai)).
 6. Click Create Bucket.
 7. Open the bucket and click Upload.
 8. Upload any two MP3 files, for example:
 - song1.mp3
 - song2.mp3
2. Apply a bucket policy to 'my-music-uploads' so only users with a specific email address (your own or a friend's) can download files from the bucket.
Hint: Use the bucket's access management settings to add an IAM user or assign a permission for a specific email.

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1. Open the bucket.
2. Go to the Permissions tab.
3. Open Bucket Policy or Access Management.
4. Grant Read (GetObject) permission only to the required IAM user or AWS account.
5. Save the policy.
Result: Only the specified user can download files from the bucket.

3. Enable server-side encryption for all objects uploaded to 'my-music-uploads', ensuring that future uploads are encrypted automatically by the cloud provider.

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1. Open the S3 bucket.
2. Select the Properties tab.
3. Scroll to Default Encryption.
4. Click Edit.
5. Enable Server-side encryption.
6. Choose:
 - SSE-S3 (Amazon S3 managed keys) (*recommended for beginners*)
7. Save the changes.

Result: All future objects uploaded to the bucket are automatically encrypted.

4. Test your bucket's security by trying to access an uploaded MP3 file from a browser where you are not logged in — take a screenshot of the error or access denied message.

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1. Copy the Object URL of one MP3 file.
2. Open a browser in Incognito/Private Mode (not logged in to AWS).
3. Paste the Object URL into the address bar.

Expected Result

- The browser should display an error similar to:
- AccessDenied

Access Denied

This confirms that the bucket is private and unauthorized users cannot access the file.

5. Use an AI tool like ChatGPT or Copilot to generate a sample bucket policy that restricts access to your bucket by IP address, then modify and apply it to 'my-music-uploads'.

Hint: Copy the AI-generated policy, review it for your cloud provider's syntax, and adjust the IP to your college's or your own network.

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- Replace YOUR-BUCKET-NAME and YOUR_PUBLIC_IP with your actual values.
- **Example**
 - If your public IP address is **103.25.120.50**, then use:
103.25.120.50/32

- After applying this policy, only requests originating from the specified IP address will be allowed to access the bucket. All other IP addresses will be denied access.

❖ **Screenshots Required**

1. S3 bucket (my-music-uploads) created.
2. Two uploaded MP3 files.
3. Default encryption (SSE-S3) enabled.
4. Browser showing **Access Denied** when accessing the object URL without logging in.
5. Bucket policy showing the IP restriction.